

Result summary

Key EDA findings

Data set size changes (cleaning/aggregation)

The exploratory analysis of the pre-processed bioactivity dataset began with 217 records. After basic data cleaning, including the removal of missing and non-numeric values, the dataset was reduced to 196 records. Further aggregation to remove duplicate entries based on identical canonical SMILES resulted in a final dataset of 158 unique compounds used for subsequent analyses

pIC50 distribution

IC50 values were converted to pIC50 to make the data much easier and safer to model and interpret. The histogram plot below shows how the pIC50 values were distributed. Those few bars around 6.5–8+ are very potent actives, but they are rare and may dominate the “active” class, making classification imbalanced and regression sensitive to outliers. It shows the dataset is dominated by low-to-moderate potency compounds, with very few high-potency actives, which drives class imbalance and risks misleading accuracy.

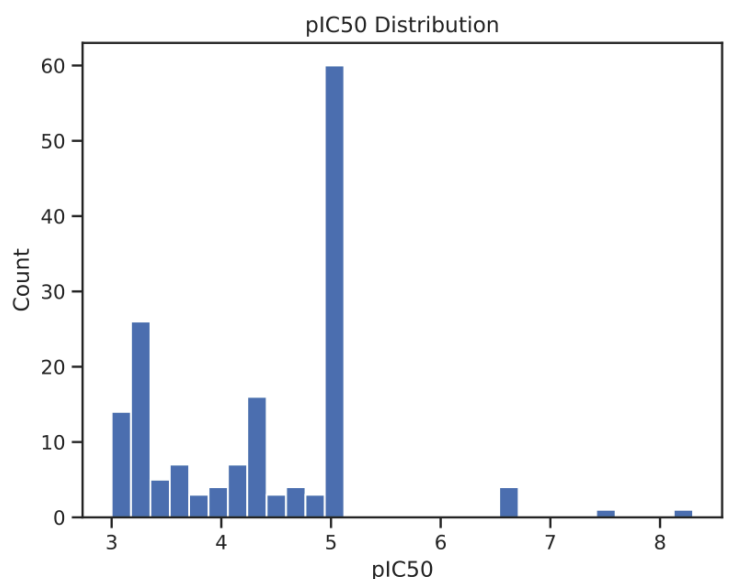


Fig 1: Histogram plot showing pIC50 count/distribution

pIC50 summary (n=158): mean ≈ 4.337 , std ≈ 0.938 , min = 3.000, median ≈ 4.333 , max ≈ 8.301 , IQR ≈ 3.338 to 5.000.

Bioactivity class balance

A bar chart was produced for class frequencies (`bioactivity_class`). Class balance is important for machine learning classification tasks because highly imbalanced dataset biases more towards the dominant class which reduces the generalization performance and leads to the misleading accuracy of the machine learning model. In this dataset we have approximately 150 inactive compounds and 10 active compounds, leading to a ratio of 1:15. This is highly imbalanced and can cause a classifier to label most compounds as inactive, achieving a deceptively high accuracy while missing true actives (low sensitivity/recall).

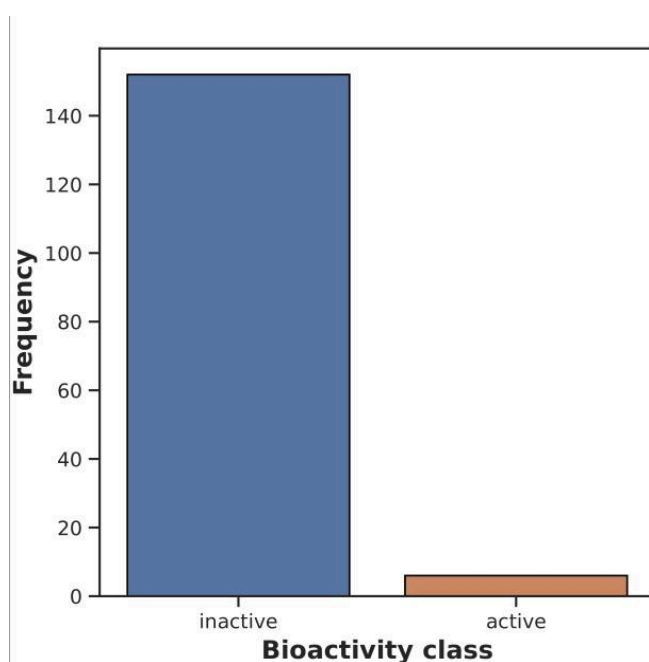


Fig 2: Bar chart showing the frequency of the bioactivity classes

Class separation

Boxplot of `pIC50` by bioactivity class shows the expected separation: actives have higher `pIC50` than inactives. The strong `pIC50` class separation indicates that bioactivity is genuinely distinguishable, which is favorable for QSAR modeling, but the small number of activities requires imbalance-aware training and careful validation.

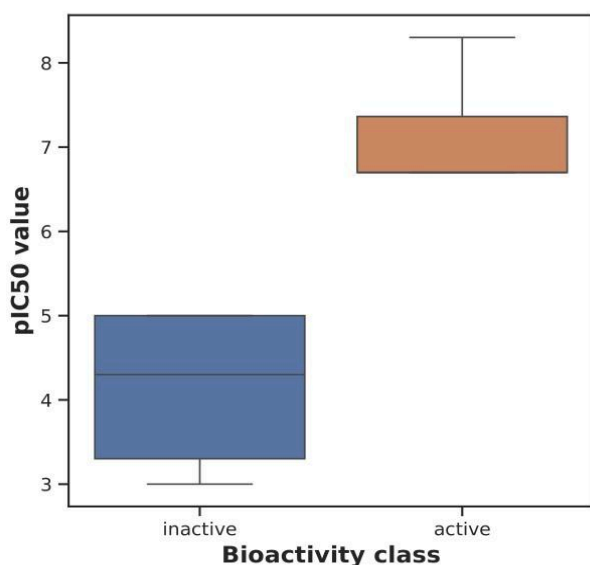


Fig 3: Boxplot showing pIC50 value by bioactivity class

The scatter plot below shows the relationship between molecular weight (MW) and lipophilicity (LogP), with bioactivity class indicated by color and pIC50 magnitude indicated by point size. It provides insight into whether simple 2D physicochemical properties can help discriminate from inactives.

The MW–LogP scatter plot shows a clear positive correlation: as MW increases, LogP generally increases. This is chemically reasonable because larger molecules often have more hydrophobic fragments. Active compounds tend to lie in a more restricted region of chemical space: MW roughly 200–500, LogP roughly –1 to ~3 and they appear to avoid extremely high MW and very high LogP regions, where many inactives are found. They occupy a constrained physicochemical region, indicating useful but limited class separation, reinforcing the need to combine simple descriptors with fingerprints for robust QSAR modeling.

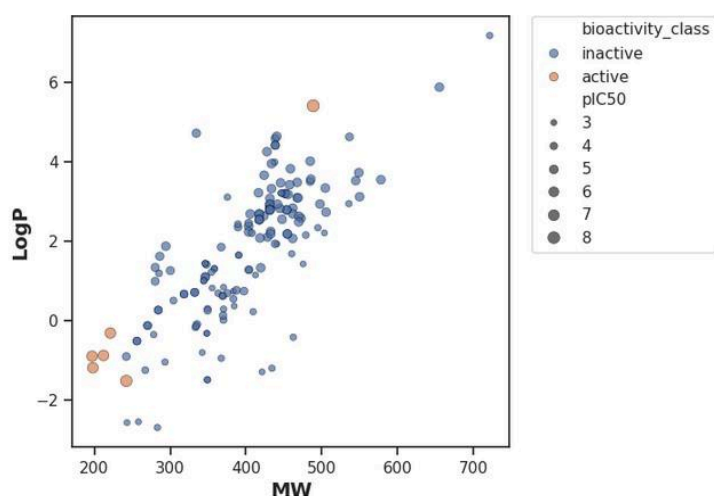


Fig 4: Scatter plot showing relationship between MW and LogP

The boxplots below compare Lipinski descriptors between inactive and active compounds, showing which simple physicochemical properties carry meaningful signals for the QSAR model and which do not.

LogP shows clear class separation and is therefore informative, suggesting that excessive lipophilicity is unfavorable for activity against this target. It should be retained as a predictive feature and may also help define the model's applicability domain. Molecular weight (MW) is highly discriminative, indicating that activity favors smaller, more compact molecules. As a result, MW will strongly influence model predictions, and the model may penalize very large compounds.

NumHAcceptors demonstrates moderate discriminatory power and is useful as a supporting descriptor, particularly when combined with MW and LogP. In contrast, NumHDonors shows no meaningful class separation, confirming the statistical tests that identified it as non-significant. On its own, it contributes little and may be down-weighted or excluded in simpler models.

In summary, the boxplots indicate that activity is associated with smaller, less lipophilic molecules that possess moderate hydrogen-bond accepting capacity, while hydrogen-bond donors contribute little to class discrimination. These findings help guide both feature selection and model interpretation.

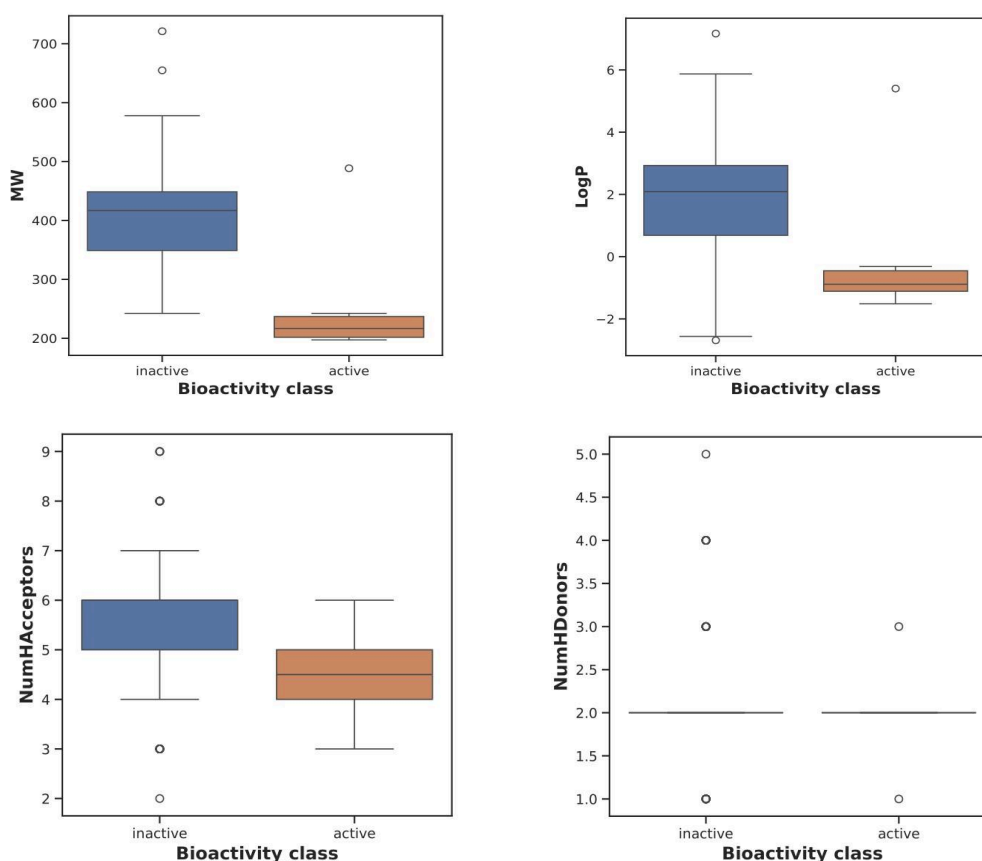


Fig 5: Box plots of lipinsky descriptors categorized by class activity

A Mann–Whitney test was carried out to statistically verify which molecular descriptors truly differentiate active from inactive compounds, thereby guiding feature selection and improving model reliability. All evaluated parameters, except NumHDonors, yielded p-values below the 0.05 threshold confirming their predictive power for distinguishing compound activity (table 1).

Table 1: Mann-Whitney Test Summary

Descriptor	Statistics	p	alpha	Interpretation
LogP	201	0.020602	0.05	Different distribution (reject H0)
MW	139	0.003985	0.05	Different distribution (reject H0)
NumHAcceptors	233	0.036697	0.05	Different distribution (reject H0)
NumHDonors	417.5	0.683222	0.05	Same distribution (fail to reject H0)
pIC50	912	1.98E-05	0.05	Different distribution (reject H0)

Calculation of 2D Descriptors

Using PaDEL (via `padel.py`), 2D molecular descriptors were computed from the canonical SMILES. In total, 1,444 2D descriptor columns were generated (excluding the auto-generated name/ID column). To remove non-informative features, constant descriptors were filtered out using a variance threshold (`VarianceThreshold = 0`), reducing the descriptor set to 1,151 unique, non-constant descriptors. The cleaned descriptor matrix was then merged with the activity metadata (`molecule_chembl_id`, `bioactivity_class`, and `pIC50`). A missing-value check found no empty values, and the final modeling dataset contained 158 compounds (rows) with 1,151 descriptor features plus the activity/ID columns.

PubChem Fingerprints Calculation

PubChem fingerprints were generated using the PaDEL-Descriptor software via `padel.py`. Each molecule was encoded as an 881-bit binary vector, where each bit represents the presence or absence of a specific chemical substructure. These

fingerprints were used as input features for QSAR model development to capture detailed structural information beyond simple physicochemical descriptors.